

Chapter 1 - Introduction to Groups

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February 25, 2026

1 Problem 1

Determine which of the following binary operations are associative:

1. The operation $*$ on $\mathbb{Z}$ defined by $a * b = a - b$.

Proof by contradiction. Assume the operation $*$ is associative. Then, $(a * b) * c = a * (b * c) \forall a, b, c \in \mathbb{Z}$.

$$\begin{aligned}(a * b) * c &= (a - b) - c = a - b - c \\ a * (b * c) &= a - (b - c) = a - b + c\end{aligned}\tag{1}$$

This is a contradiction, as for any fixed a and b , $\exists c : a - b - c \neq a - b + c$. Take any $c \neq 0$, for example. Thus, the operation $*$ is **not associative**.

2. The operation $*$ on $\mathbb{R}$ defined by $a * b = a + b + ab$.

For $a, b, c \in \mathbb{R}$,

$$\begin{aligned}(a * b) * c &= (a + b + ab) * c \\ &= (a + b + ab) + c + (a + b + ab)c \\ &= a + b + ab + c + ac + bc + abc \\ &= a + b + c + bc + ab + ac + abc \\ &= a + (b + c + bc) + a(b + c + bc) \\ &= a * (b + c + bc) \\ &= a * (b * c)\end{aligned}\tag{2}$$

$\forall a, b, c \in \mathbb{R}$, $(a * b) * c = a * (b * c)$. Thus, the operation $*$ is **associative**.

Version 2 - The Imposter Additive Operation

Notice that $a + b + ab = (a + 1)(b + 1) - 1$ and define the map $\varphi : (\mathbb{R}, *) \rightarrow (\mathbb{R}^\times, \circ)$ as $\phi(a) = a + 1$, or a linear translation over $\mathbb{R}$.

$$\begin{aligned}
\phi(a * b) &= 1 + (a * b) \\
&= 1 + (a + b + ab) \\
&= (a + 1)(b + 1) \\
&= \phi(a) \circ \phi(b)
\end{aligned} \tag{3}$$

So, φ is a homomorphism. It must now be determined if φ is bijective. Proof by contradiction is used to show injection. Assume $a, b \in \mathbb{R} : a \neq b$ map to the same element $s \in \mathbb{R}$ in the range.

$$\begin{aligned}
\varphi(a) &= \varphi(b) = s \\
a + 1 &= b + 1 = s \\
a &= b = s - 1
\end{aligned} \tag{4}$$

This shows $a = b$, reaching a contradiction, and indicating that the assumption of $a \neq b$ is false. Proof by construction is used to prove surjectivity. For all elements $c \in (\mathbb{R}, \circ)$, there exists an element that maps to it, namely a left translation by 1. Because φ is bijective, it is an isomorphic map, and there exists an inverse $(\mathbb{R}^\times, \circ) \rightarrow (\mathbb{R}, *)$ defined as $\phi^{-1}(b) = b - 1$.

$$\begin{aligned}
(a * b) * c &= \varphi^{-1} [((\varphi(a * b)) \circ \varphi(c))] \\
&= \varphi^{-1} [((\varphi(a) \circ \varphi(b)) \circ \varphi(c))] \\
&= \varphi^{-1} [(\varphi(a) \circ (\varphi(b) \circ \varphi(c)))] \\
&= \varphi^{-1} [(\varphi(a) \circ \varphi(b * c))] \\
&= \varphi^{-1} [(\varphi(a) * \varphi(b * c))] \\
&= \varphi^{-1} [(\varphi(a) * \varphi(b * c))] \\
&\quad a * (b * c)
\end{aligned} \tag{5}$$

$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in \mathbb{R} \quad \text{By associative property of multiplication.} \tag{6}$$

2 Problem 5

Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

The left and right cancellation laws must hold for a group G .

$$\begin{aligned}
au = av &\implies u = v \quad \forall a, u, v \in G \quad \text{By left cancellation.} \\
ua = va &\implies u = v \quad \forall a, u, v \in G \quad \text{By right cancellation.}
\end{aligned} \tag{7}$$

However, for $\mathbb{Z}/n\mathbb{Z}$, $\bar{0} * \bar{a} = \bar{0} * \bar{b}$, even when $a \neq b$. Thus, $\mathbb{Z}/n\mathbb{Z}$ for $n > 1$ is **not a group**.